Release Notes

CADP internal QA build release notes — generated from OpenProject. Not the in-app Release Notes feature used by tenants.

1. Release Overview

Field	Details
Product Name	**CADP Workspace**
Release Version	**Release 1.0.0 (27)**
Release Date	**October 28, 2026**
Release Type	**Feature & Bug Fix Release**
Prepared By	**Development Team**
Approved By	**Ganagavathy**
Environment	**QA**

2. Release Summary

This release of CADP Workspace Version 1.0.0 (27) delivers 0 bug fixes, and 1 new feature across the platform. Highlights include Fetch Records node: palette, config panel, test run, Run for, @ picker (Phases 5–6). It also includes a required configuration change (see Configuration Changes) and API changes (see API Changes). Refer to the Known Issues section for current limitations.

3. New Features

Feature ID	Feature Name	Description	Impact
	Fetch Records node: palette, config panel, test run, Run for, @ picker (Phases 5–6)	Workflow Builder support for the backend `fetch-record-node`; (`services/cadp_workflow_backend_nodejs/docs/changes/fetch-record-node.md`).; **Node registration:** `fetch-record-node` in the `WorkflowNodeType` union, a; `NODE_DEFINITIONS` entry (category action, "Fetch Records"; `defaultConfig` copied key for; key from the backend catalog), and the Database icon in the 3 icon maps.; **Branch handles:** `found`, `empty`, plus `error` when Continue On Error is on; (`branchUtils.getNodeBranchDefinitions`). These are the labels the engine routes on.; **Config panel** (`FetchRecordConfigPanel`, routed by type in `NodeConfigPanel`, the same; pattern as Approval Gate). Sections:; 1. Source: module, feature, optional Data Table.; 2. Filter: nested ALL/ANY groups with `@`/`{{}}` token values; skip-blank and; include-deleted toggles.; 3. Result: single / list / count / exists / aggregate; fields to return; group by,; aggregates and having; sort; limit.; 4. Output: output variable (validated `^\w+$`), output rows field, when nothing is found,; continue on error, timeout.; **Downstream tokens:**; `useUpstreamSources` derives a Fetch node's outputs from its returned fields and; aggregate aliases: `{{<nodeId>.record.X}}` / `{{<nodeId>.rows[0].X}}`, count, found.; It no longer lists the fetched feature as an upstream *trigger* feature (its values do; not come from the trigger record).; `useFieldTagging` offers upstream Fetch outputs even when the caller passes none, and; `{{row.X}}` entries ("Fetch row" group, listed first) when the step's Records Field equals; a Fetch node's Output Rows Field.; **Run for** (`RunForSelector`, shown above Create / Update / Upsert / Delete):; Options are "This trigger record" or "Each row from: <Fetch node>", and it writes; `recordsField`.; Switching back to the trigger record asks for confirmation when `{{row.…}}` tokens are; used.	

4. Enhancements / Improvements

Enhancement ID	Module	Description	Benefit
	None	No enhancements recorded yet for this release.	N/A

5. Bug Fixes

Bug ID	Description
—	No fixed bugs recorded yet for this release.

6. Configuration Changes

Configuration	Description	Impact
Vault secret names for AI builder / LLM gateway switched to kebab-case	`runtime-bootstrap.js` now resolves `AI_BUILDER_BASE_URL`, `AI_BUILDER_USERNAME`,; `AI_BUILDER_PASSWORD`, `AI_BUILDER_HIERARCHY`, `LLM_GATEWAY_URL` and `LLM_GATEWAY_API_KEY` from the; vault secrets `ai-builder-base-url`, `ai-builder-username`, `ai-builder-password`,; `ai-builder-hierarchy`, `llm-gateway-url` and `llm-gateway-api-key` (previously the UPPER_SNAKE names).; Env var names consumed by the code are unchanged.	The vault keys were renamed to kebab-case to match the other secrets.

Configuration	Description	Impact
Faster workspace-access lookup and perf measurement in RBAC	New roles index `{ tenantId: 1, workspaceId: 1, roleId: 1 }`. The only existing index; starts with `roleName`, but the role lookups made on every authenticated request; (`getRolesByNamesAndTenantId`) and by `GET /users/workspace`; (`getWorkspaceIdsWithDirectRoles`, `getWorkspaceIdsWithGroupRoles`) filter on; tenantId/workspaceId/roleId and never on roleName, so they could not use it.; Mongoose `autoIndex` builds it on each tenant DB the first time the Role model is; bound there after deploy (non-unique, no data changes).; `getWorkspaceIdsByTenant` (`GET /users/workspace`) now selects only the user fields; it reads instead of the whole user document. Response unchanged.; Ported master's perf measurement (`src/utils/perf-metrics.js`,; `src/middleware/perf-timing.middleware.js`): per-request timing with Mongo op; count/time, `[perf] slow mongo command` (≥ `PERF_SLOW_MONGO_MS`, default 100),; `[perf] slow outgoing http` (≥ `PERF_SLOW_HTTP_MS`, default 500), and a periodic; per-route p50/p95/p99 summary. Same `PERF_*` env vars as master. Because every; browser request passes through RBAC, `[perf] request` lines default to requests; ≥ 500 ms (`PERF_LOG_MIN_MS`). Requests forwarded by the proxy routers group by; their id-normalised path, tagged `[proxied]`. Log lines carry field names only,; never values, bodies or tokens.	master logged `[perf] slow outgoing http` for; `GET localhost:3001/rbac/users/workspace` at 2,890 ms on; `GET /master/workspaces/workspacesList` in UAT, and RBAC had no measurement to show; which step was slow.
LLM gateway URL resolved from vault	`runtime-bootstrap.js` now maps `LLM_GATEWAY_URL` from the vault secret `llm-gateway-url`.; Optional; falls back to the `.env` value / built-in default as before. The API key still comes from; each saved LLM integration.	Aligns with master backend's vault key naming.

Configuration	Description	Impact
Perf measurement in client-ws	Ported master's perf measurement (`src/utils/perf-metrics.js`,; `src/middleware/perf-timing.middleware.js`): per-request timing grouped by Express; route with Mongo op count/time, `[perf] slow mongo command`; (≥ `PERF_SLOW_MONGO_MS`, default 100), `[perf] slow outgoing http`; (≥ `PERF_SLOW_HTTP_MS`, default 500) and a periodic per-route p50/p95/p99 summary.; Same `PERF_*` env vars as master. Most traffic here is master's service-to-service; role checks, so `[perf] request` lines default to requests ≥ 500 ms; (`PERF_LOG_MIN_MS`). Log lines carry route templates, collection and field names; only, never values, bodies or tokens.	master could see that client-ws role checks were slow; (`POST /role/view/features`, 1,199 ms in UAT) but client-ws had no measurement to; show which step was slow.
AI builder credentials resolved from vault	`runtime-bootstrap.js` now maps `AI_BUILDER_BASE_URL`, `AI_BUILDER_USERNAME` and; `AI_BUILDER_PASSWORD` from the vault secrets `ai-builder-base-url`, `ai-builder-username` and; `ai-builder-password`. Optional; `.env` values still apply when the secret is absent.	Aligns with master backend so AI workflow generation reads credentials from the vault; instead of plaintext env files (addresses the follow-up above).

7. API Changes

API Name	Type	Description
All external-database endpoints moved to cadp_dbsync_backend_nodejs		Follow-up to the Database Sync move below: the rest of the external-database; code moved to `cadp_dbsync_backend_nodejs` as well, so master no longer contains any.; Endpoints: `/external-db/explore`, `/query`, `/aggregate`, `/export`, `/bulkdataFetch`,; `/distinct-values`, `/select`, `/select/validate` (plus `/test-query`, `/test-connection`; moved earlier).; Code: `external-db-explorer.controller.js`, `external-db-select.controller.js`, their; routes, `external-db-{aggregate,distinct,explorer,export,page,select}.service.js`, and; the whole `src/service/external-db/` folder including the SQL core (`core/`).; Tests: external-db-page, external-db-select, external-db-statement-timeout and the nine; sql-core-* suites moved with the code.; Master keeps the shared helpers that other code uses: `constants/dacFilterOperators.js`; (extraction DTO), `utils/filterGroupUtils.js`, `communicators/client.ws.service.js`.; `data-archival/rules/readOnly.rules.js` still lists `/external-db/*` as read-only routes;; those entries no longer match anything in master and are harmless.; The external-DB change logs (external-db-explorer.md, external-db-sql-core.md,; external-db-dac.md) were copied to the dbsync repo; the copies here are marked moved.

API Name	Type	Description
Database Sync moved out of master into cadp_dbsync_backend_nodejs		Database Sync now runs in its own service, `cadp_dbsync_backend_nodejs`.; Master no longer runs syncs or serves the query builder's test endpoints.; **Moved out of master** (now in the dbsync repo): the scheduled run; (`GET /database-sync/run/:featureId`), scheduler registration (create/recreate/remove; the notification-service task), `bulkSyncFeatureData`, `reconcileFullSyncDeletes`,; `resolveImportFieldMetadataForFeature`, `ensureSyncMatchIndex`,; `external-db-query.controller/service/routes` (`POST /external-db/test-query`,; `POST /external-db/test-connection`), `external-db/json-path-explode.js`,; `external-db/row-group-in-memory.js`, and the notification-service connector.; The import/auto-ID helpers those methods call stay in master (regular imports use; them); dbsync carries its own copy of the 48 methods the sync needs.; **Master still** validates and saves the feature's `databaseSync` config. After a; save that touched config, and after a feature delete, it now calls; `POST {DBSYNC_SERVICE_URL}/database-sync/schedule/:featureId` (new; `communicators/dbsync.service.js`); the dbsync service owns the scheduler task and; stores `schedulerTaskId`/`schedulerTriggerId` on the feature. Still fire-and-forget.; The frontend calls `/dbsync/external-db/test-query` through the RBAC gateway's new; `/dbsync` proxy.; The dbsync service adds a per-feature run lock (one run per feature across all its; pods) and a 30-minute scheduler timeout for sync actions. Overlapping runs were; visible in UAT logs: a run longer than the scheduler's 5-minute HTTP timeout was; retried while still running.
Workspace list: page size capped at 100, version lookup scoped to the page		`GET /workspaces/workspacesList` now caps `limit` at 100 (minimum 1, default; 10) and treats an invalid or < 1 `page` as 1. The response's `pagination.limit`; reports the capped value. The active-version lookup now reads only the page's; workspaces and only `{ workspaceId, version }`, instead of the full active-version; record of every workspace in the tenant. `getAllActiveVersions(workspaceIds)` keeps; its old behaviour when called without ids; `findLatestActiveByFilter` takes an; optional projection. Also fixed: with `limit=1` and a pinned workspace, page 1; ran `.limit(0)`, which Mongo treats as "no limit", so it returned every workspace.; That query is now skipped.

8. Known Issues

Issue ID	Description	Workaround
	Role Management Change Effect	Because some menu structures have changed, certain features may return a 403 status code. To resolve this, edit the role, select the platform feature, and save the role again.
	Scheduler with Workflow Not Working	Use the conventional feature-wise approach wherever workflow scheduling is required.
	Formula Builder UI – Field Insertion Through Typing	This works in current fields. For external feature fields, it won't work, as there could be fields with same names from multiple features.
	Formula Builder – AI-Based Formula Writing/Editing Not Implemented	Use manual formula entry instead.
364	Organization Details → Basic Info: Website URL accepts invalid URL formats without validation.	Expected: Validate the URL format and show an error message for invalid values, such as www.example or example@123.www.example
	Builder Tool Not Available on Mobile Web	The builder tool is not supported on mobile web. This is implemented as expected — builder functionality is restricted to desktop browsers by design. It should be available on Tab though.
	Mobile – Formula Field Not Working in Tables & Charts	Formula fields are not evaluated correctly when used inside Tables and Charts on mobile. This is under review.

9. What to Test

- Login – Login, Sign In with Google (have to be configured in external providers in Tenants of metadata), Sign In with Microsoft (have to be configured in external providers in Tenants of metadata)
- **Create Application**
    1. Start from Scratch
    2. Choose/Create Vertical Template

- **AI-Assisted Creation & Ask FAQ** (Cavin's AI assistant — the AI launcher icon)
    1. **AI App Creation**
        a. From the workspace (any page other than Modules Management / Features Management / Builder Tool), open Cavin's AI and select the **Create** tab.
        b. Verify the helper line reads "Create builds a new application from your description".
        c. Describe the application to build and send. Verify creation progress is shown and "Application created from AI." appears on completion.
        d. Verify the generated application, its modules and features are listed and can be opened in the workspace.
        e. If the AI asks clarifying questions, answer them and verify creation continues (no success message is shown until something is actually created).
    2. **AI Module Creation**
        a. Open an application's **Modules Management** page, open Cavin's AI and select the **Create** tab.
        b. Verify the helper line reads "Create builds a new module here".
        c. Describe the module to build and send. Verify "Created N module(s) from AI." appears and the module list refreshes with the new module(s).
        d. Open the created module and verify it can be edited further.
    3. **AI Feature Creation**
        a. Open a module's **Features Management** page (or the Builder Tool), open Cavin's AI and select the **Create** tab.
        b. Verify the helper line reads "Create builds a new feature here".
        c. Describe the feature to build and send. Verify "Created N feature(s) from AI." appears and the feature list refreshes.
        d. From the Builder Tool, verify you are taken to the first created feature in the builder.
        e. Verify the generated form has appropriate components/fields, then publish it and verify records can be created, viewed and edited.
    4. **Create / Ask mode behaviour**
        a. In the Create tab, type a question instead of a build request. Verify nothing is created and an "Ask this in Ask mode" suggestion is offered; clicking it moves the question to the Ask tab.
        b. With a user who has only AI Builder or only AI Ask access, verify only that tab is available; with no create access, verify past conversations are read-only.
        c. Verify conversation history can be searched and reopened for each mode.

5. **AI Ask FAQ**

   Scope: AI Ask FAQ is trained only on the following platform documentation. Questions must be asked on these topics only:

   - CADP User Login Guide
   - CADP Functional Specifications
   - CADP Erudite Management Complete
   - CADP Workspace Guide
   - CADP User Preferences Guide
   - CADP Role Management
   - AI FAQ Guide Transactions Runtime
   - CADP Login Complete
   - CADP Erudite Management Complete Guide
   - CADP Features Guide
   - CADP Modules Guide
   - CADP Platform Architecture Report
   - CADP Templates Integrations Workflows Basics Guide
   - CADP Features And Submodules Guide
   - CADP Organization Guide
   - CADP Builder Tool Guide

   a. Open Cavin's AI, select the **Ask** tab and ask a question about platform configuration on one of the trained topics above.

   b. Verify a relevant and accurate answer is returned.

   c. Ask a follow-up or ambiguous question and verify it is handled gracefully.

   d. Ask a question outside the trained topics and verify AI Ask FAQ declines or returns an appropriate out-of-scope fallback rather than a fabricated answer.

   e. Verify Ask mode changes nothing in the workspace (it only answers).

- **User Management**

  1. Create/Manage Users
  2. Create/Manage Groups
  3. Import/Export Users
  4. Archive/Unarchive Users
  5. Locked Users
  6. Waiting Approval
  7. Please update existing users in each workspace.
  8. In the Users tab, when creating a new user, the Approval Type default is No Approval.

- **User Preferences**
    1. Profile update – first name, last name updating, profile picture updating.
    2. Change password
    3. Locale and Date-time updating
    4. Interface
        i. Theme change, color change. Color change here will override the brand color for that specific user of that workspace only.
        ii. Records per page is reflecting.
        iii. Language (currently only English accepted)
    5. Customization – module and features are visible based on visibility setting, quick access for that enabled feature is redirecting to feature transactions (records) page.
    6. Privacy & Security
    7. Miscellaneous – Default landing page, preferred currency, auto-save drafts, save interval
    8. Preferred user workspace
    9. Reset Preferences
- **Groups**
    1. Archive/Unarchive

- **Roles**
    1. Platform Roles
    2. Archive/Unarchive
    3. Workspace Role
    4. Feature Access Level
    5. Data Access Level
    6. User or Group Role Mapping
- **Workspace Management**
    1. Create Workspace
    2. Clone Workspace
    3. Archive Workspace
    4. Workspace Settings
- **Organization Management**
    1. Organization Details
    2. Organization Address
    3. Branding Details
- **Module/Feature Management**
    1. Create/Manage Module
    2. Create/Manage Features
    3. Design components for features
    4. Features and Modules can be published
    5. Audit Logs (Platform & Workspace)
    6. Version Logs (Workspace & Workflow)
    7. Staging
    8. Feature Card and Module Card Customization
- **Notification & Alerts**
    1. Email Notification & Templates – Create template and use template to send notification
    2. SMS Notification & Templates – Create template and use template to send notification
    3. WhatsApp Notification & Templates – Create template and use template to send notification
    4. In-App Notification & Templates – Create template and use template to send notification
    5. Push Notification & Templates – Create template and use template to send notification

- **Erudite Management**
  1. **Configuration (prerequisite)**
     a. Enable Erudite Management in Role Management before testing.
     b. Verify users without the role cannot access the Erudite Management module.
  2. **FAQs**
     a. Create a new FAQ with a question, answer, and category (Account, Billing, General, Technical).
     b. Verify the FAQ appears in the FAQ list after saving.
     c. Enable "Display this on the dashboard" and verify the FAQ is visible on the dashboard.
     d. Edit an existing FAQ and verify changes are saved correctly.
     e. Delete a FAQ and verify it is removed from the list.
     f. Search for a FAQ by keyword and verify correct results are returned.
  3. **Documentation**
     a. Upload a document (PDF or DOC, max 5 MB) with a name, description, and category.
     b. Verify the document appears in the Documentation list after saving.
     c. Attempt to upload a file exceeding 5 MB and verify a validation error is shown.
     d. Attempt to upload an unsupported file format and verify it is rejected.
     e. Enable "Display this on the dashboard" and verify the document is visible on the dashboard.
     f. Edit document details and verify changes are saved correctly.
     g. Delete a document and verify it is removed from the list.
  4. **Tutorial Videos**
     a. Upload a video (MP4, MOV, or MKV, max 50 MB) with a name, description, and category.
     b. Verify the video appears in the Tutorial Videos list after saving.
     c. Attempt to upload a file exceeding 50 MB and verify a validation error is shown.
     d. Attempt to upload an unsupported file format and verify it is rejected.
     e. Enable "Display this on the dashboard" and verify the video is visible on the dashboard.
     f. Edit video details and verify changes are saved correctly.
     g. Delete a video and verify it is removed from the list.
  5. **End User Access**
     a. Verify end users can browse and read FAQs, documentation, and tutorial videos.
     b. Verify items marked "Display on dashboard" appear on the user's dashboard.
     c. Verify items not marked for dashboard display do not appear on the dashboard.

- **Components**
    1. Layouts
        1. Column
        2. Section (various nested scenarios)
    2. Data Management (API Integration, Device Integration)
    3. Input Components (Phone Number, Password, Dropdown with Feature as Field Value)
    4. Specialized Input & Display
    5. Display and Interaction
    6. Media and Files
    7. Forms and Validation
    8. Utility Components
    9. Data Config Component
    10. Hyper Link Component
    11. Favorite Component
    12. OCR Component
        a. Add the OCR component to a feature form and upload an image or document.
        b. General text extraction → verify the text is extracted from the uploaded image/document and displayed within the selected output textarea
        c. Verify the extracted text is saved correctly with the record and is visible on View/Edit.
    13. Analytics & Report Components
        1. Chart Component
        2. Gantt Chart Component
        3. KPI Card Component
    14. Form UI
        1. Table component and Table level formulas, Summary row formulas
    15. Reusable Component
- **Page Builder**
    1. Create a new page and verify it is saved and listed.
    2. Drag-and-drop layouts (columns, sections) and reusable components onto the page canvas.
    3. Configure component properties and verify changes reflect on the page.
    4. Reorder, move, and delete components on the canvas.
    5. Edit an existing page and verify changes are persisted.
    6. Delete a page and verify it is removed from the list.
    7. Publish the page and verify it renders correctly at runtime.
    8. Verify the page is responsive across mobile, tablet, and desktop.
    9. Verify role-based access — only authorized users can build/edit pages.

- **Workflow Management**
    1. Visual canvas for creating workflows
    2. Drag-and-drop node placement
    3. Node connection with visual links
    4. Canvas zoom and pan controls
    5. Grid snap for alignment
    6. Create new workflows
    7. Edit existing workflows
    8. Delete workflows
    9. List/view all workflows
- **Workflow Nodes – Control Flow**
    1. Start Node (workflow trigger point)
    2. End Node (workflow termination)
    3. Condition/Decision Node (if-else logic)
- **Workflow Nodes – Communication**
    1. Email Notification Node
    2. SMS Notification Node
    3. User Task/Approval Node
    4. System Notification Node
- **Workflow Monitoring**
    1. Execution history logs
    2. Workflow instance status tracking
    3. Node-level execution details
    4. Error logs and stack traces
    5. Approval node changes for user selection
    6. Added SLA and escalation reminder for approvals
    7. Added parallel workflow approval node
- **Printing Templates**
    1. QR Code – Generate and Print
    2. Barcode – Generate and Print
    3. Invoice – Generate and Print
- **Role-Based Dashboards**
    1. Role-based dashboard template creation
    2. Dashboard rendering based on created template

- **Scheduler & Task Scheduler**
    1. Create and use Schedule
    2. Create and use Task Schedule
    3. Event-based task schedule
    4. Schedule-based task schedule
    5. Schedule a notification
    6. SMS
    7. Mail
    8. WhatsApp
    9. Push
    10. In-App
    11. Schedule a report send (PDF) through mail from a configured email template.
- **Formula Builder Functionality**
    1. Same Feature
        a. Arithmetic
        b. Logical
        c. Comparison
        d. Text
        e. Math
        f. Date
        g. Conditional
        h. Miscellaneous
        i. Aggregate → with filter conditions and without filter conditions.
    2. Cross Feature
        a. Arithmetic
        b. Logical
        c. Comparison
        d. Text
        e. Math
        f. Date
        g. Conditional
        h. Miscellaneous
        i. Aggregate → with filter conditions and without filter conditions.
    3. Incompatible datatype operations, for example, arithmetic on a text input field, should show a validation error.
    4. Unrecognized fields, functions, or operators should show a validation error.
    5. If external fields are called without where() conditions, it should show MULTIPLE RECORDS FOUND ERROR in preview for non-aggregate functions/operations.
    6. Variables are not working. Review which variables should be displayed in the modal.

- **Formula Builder UI**
  1. On clicking a feature from the feature list, fields in the selected feature should be displayed, except for the current field, to prevent self-call loops.
  2. On selecting external feature fields, where() should appear in the modal.
  3. On selecting where(), the modal should automatically switch to the Feature and Fields tab.
  4. On hovering over a field, field information such as module name and feature name should be displayed.
  5. The Clear button or backspace should work as follows:
     i. First click of clear/backspace will select the token.
     ii. Next click of clear/backspace will clear that token.
  6. Empty formulas should not be saved.
  7. Formulas should not be saved without validation.

- **Formula Builder – Helper Tools**
  1. **Cheat Sheet** (**?** icon)
     a. Open the Cheat Sheet and verify functions are grouped by category (Lookup, Nested Lookup, Aggregation, Conditional, Math, String, Date).
     b. Search for a function and verify the list filters correctly.
     c. Click the copy icon on an example and verify it is copied to the clipboard.
     d. Press Esc and verify the modal closes.
  2. **Examples**
     a. Open the Examples drawer and verify 50 scenarios across all categories are listed.
     b. Filter by a category and search; verify results update.
     c. Click "Use This Pattern" and verify the formula is inserted into the editor.
  3. **Test Formula (Playground)**
     a. For a current-feature calculation (e.g. *{number_field} * 5*), open Test Formula, verify a type-aware input appears per field (number/date/true-false/text), enter values, run, and verify the correct result, type, and (where relevant) matched-record count.
     b. For a lookup formula (uses *.where()* / external feature fields), verify NO value-input form is shown and a "Lookup formula – test against a real record" message appears instead.
     c. For an aggregate formula (SUM/COUNT/AVG/MIN/MAX of records), verify the "Aggregate formula – test against a real record" message is shown and the form is not available.
     d. Verify the Test Formula button is disabled when the formula is empty.
  4. **Template Library**
     a. Open the Template Library and verify the **Most Used** tab lists the 20 platform built-in templates (e.g. GST, TDS, net salary, lookups, aggregations).
     b. Search and filter by category; verify results update.
     c. Click "Use This Template" and verify the formula is inserted into the editor.
     d. With a formula in the editor, click "Save current formula as template", fill title/category/description, save, and verify it appears under **My Templates**.
     e. Delete a saved (non built-in) template and verify it is removed. Verify built-in templates cannot be deleted.
  5. **Formula Wizard**
     a. **Lookup mode:** choose "Lookup a value", select a source feature, select the field to return, add a match condition (verify the "Right side is a field reference" option shows a dropdown of the current feature's fields), check the live preview, and Insert.
     b. **Aggregate mode:** choose "Aggregate records", select feature/field, optionally add one filter condition, choose the function (SUM/COUNT/AVG/MIN/MAX), and Insert.
     c. Verify the formula preview only appears after a return field is chosen, and that going Back keeps your previous selections.
     d. After inserting, hover over the tokens in the editor and verify the correct Module name and Feature name are shown for each (including the source feature for the looked-up field).

6. **Smart Validation Messages**
    a. Enter an invalid formula (e.g. unknown function, missing where(), wrong argument count) and click Validate Expression; verify the error includes a helpful suggestion and an example, not just a generic message.

- **Integrations**
    1. Step 1: Go to the Integration Type field and select "Third-Party Software".

       Expected Result: A new dropdown field called Select Type should appear.
    2. Step 2: Check the options inside the Select Type dropdown.

       Expected Result: It should contain options including "Third-Party API" and "Google Drive".
    3. Step 3: Choose "Third-Party API" from the dropdown.

       Expected Result: The configuration fields specific to the Third-Party API should be displayed.
    4. Step 4: Change the selection to "Google Drive".

       Expected Result: The system should dynamically display the Google Drive configuration fields: Client ID, Client Secret, and Refresh Token.

This build — additional test focus:

- Roles assigned by their record ID now load correctly — `getRolesByNamesAndTenantId` matched a user's role refs against `roleId` and; a misspelt `__id` field that no role has, so a role referenced by its Mongo `_id`; never matched. It now matches `roleId` or `_id` (for refs that are 24-hex ObjectIds),; the same way the workspace-access checks in `user.workspace.service.js` already do.; Empty/missing refs skip the query and return `[]`, the same result as before.

10. What to Test on Mobile

- Login – Normal Login
- Login – SSO (Sign In with Google, Sign In with Microsoft)
- Workspace Switch
- Module List
- Feature List
- Components
- Feature Data – Create, View, Edit, Clone, Delete
- Dashboard
- Integration – Weighing Machine, Third-Party API (Budgie, SAP), and Payment Gateways (Stripe, Razorpay, and PayU)
- Theme
- Push Notification
- Workflow – Approve and Reject
- Charts – Feature Data
- Data Table – Feature Data
- Security – Prevent copy, paste, cut, and text selection